# CAS (Heimdall) – system functional description

## Inlet system: water intake and pen level control

ScaleAQ Automation Document CAS-FD-10, rev. B

October 23, 2026

**Abstract**

How the inlet system of the CAS (Heimdall) closed fish pen works and how the control system controls it. Three Framo inlet pumps fill the pen with sea water; the Heimdall PLC holds the water inside the pen a few centimetres above the sea outside (the *over-height*) with one common pump speed, and chooses how many pumps run. The document covers every instrument and actuator, the control principle, sequences, the pump state machine, interlocks, alarms and faults. Undecided points are marked OIn and listed in Section 6.

## Contents

**1 Introduction** **1**
1.1 Purpose and readers . . . 1
1.2 Scope . . . 2
1.3 Sources . . . 2
1.4 Abbreviations and terms . . . 3
1.5 Revision note . . . 3

**2 Block diagram description** **4**
2.1 The process in brief . . . 4
2.2 Physical units and networks . . . 4
2.3 Simplified process and instrument diagram . . . 4

**3 Functional diagram description** **6**
3.1 Top-level functions . . . 6
3.2 Level control loop LIC01 . . . 7
3.3 Pen start and stop sequence . . . 8
3.4 Pump count and rotation . . . 9
3.5 Framo pump state machine . . . 10
3.6 Heartbeat . . . 11
3.7 Operating modes . . . 11

**4 Functional requirements** **11**
4.1 Processes . . . 11
4.2 Inputs . . . 12
4.3 Outputs . . . 12
4.4 Startup and shutdown . . . 12
4.5 Fault identification and recovery . . . 13
4.6 Interlocks . . . 14
4.7 Communication . . . 14

**5 Non-functional requirements** **15**
5.1 Control law and tuning . . . 15
5.2 Cycle times and tasks . . . 16
5.3 Signal resolution and filtering . . . 16
5.4 Alarm functions . . . 17
5.5 Parameters . . . 18
5.6 Interfaces: instruments and actuators . 19
5.7 Interfaces: OPC UA signals to and from the Framo controllers . . . 20
5.8 HMI operator functions . . . 22

**6 Open issues and decisions** **22**
6.1 Open issues . . . 22
6.2 Decisions . . . 24

# 1 Introduction

## 1.1 Purpose and readers

The document is the system functional specification of the inlet system in the sense of [COO] §14.3: a description of the physical system (*block diagram description*), of what the control system does with it (*functional diagram description*) and of the functional and non-functional requirements. It is written for

- **engineers and the project team**, who implement, test and commission the PLC program;

- **operators**, who need to know what the system does on its own and what they must do;
- **suppliers** (Framo for the inlet pumps), who must agree the interface and the behaviour at the boundary.

[FD] (Norwegian) is authoritative on the *intent*. This document gives the intent in English, shows what the PLC program [PLC] implements today (main, 2026-10-22) and where they differ.

## 1.2 Scope

**In scope:** the water intake from the three intakes to the pen (Framo inlet pumps PM-31/32/33, their local controllers and vacuum systems as seen from the PLC), the level measurement inside and outside the pen, the level controller LIC01, the start and stop of the pen and of each pump, the communication with the Framo controllers and the related alarms.

**Out of scope** (separate FDs): the oxygen system (side stream, oxy pumps PU41–43, OIC01), the outlet dampers and their hydraulics, the bilge system in the inlet boxes, and the power distribution. They appear here only where they interface with the inlet system: the oxygenation permit per line, the level setpoint per operating mode from the outlet, and interlocks from the inlet boxes.

## 1.3 Sources

Table 1: Sources. The IDs are used in the text.

| ID | Document | Revision / date |
|---|---|---|
| [FD] | P-15169-CAS-FD-01 *Funksjonsbeskrivelse CAS*, §3.2.1 Vanninntak, §3.2.2 level control, tables 5–10 (Norwegian) | rev. B, 13.05.2026 |
| [AL] | P-15169-CAS-FD-02 *Alarmliste CAS Heimdall* | rev. A |
| [FRAMO] | Framo, *Spesifikasjon: kontrollsystem (prototype)* 0000_CAS_SPEC_CTRLSYS, signal list and state table, with ScaleAQ review comments | rev. B, 08.10.2025 |
| [FIL] | Framo interface list (early signal list) | 01.09.2025 |
| [PID] | CAS P&ID and loop diagrams: CAS-PID-100 inlet, -400 pen, -600 utilities, CAS-LAY-700 network, CAS-LD-801 loop LT01, CAS-PID-950 open points | rev. A, 2026-10-16 |
| [DES] | CAS design basis (System Modelling), generated from design_basis.toml | v2.3.12 |
| [PDS] | CAS program design (dataflow of the level loop, start command path) | 2026-10-15 |
| [PLC] | Heimdall PLC program (CODESYS, application PFC200_Sim): FB.Inlet.InletControl, InletSupervisor, InletSensors, FB.Pen.PenControl, PRG.FramoDataProcessing, GVL ConfigValues | main, 2026-10-22 |
| [EL] | Electrical drawings 15169-792.001 (PLC/RIO1 cabinet), 792.002 (RIO2/3 cabinet), 874.001 (Framo pump cabinet) | 792.001 rev. 1, 792.002 rev. 0 |
| [COO] | J. Cooling, *Software Engineering for Real-Time Systems*, 2020, §14.3 System functional specifications | 2020 |

## 1.4 Abbreviations and terms

| Term | Meaning |
|---|---|
| CAS | Closed Aquaculture System; Heimdall is the control system of the CAS pen |
| Over-height, $\Delta H$ | Water level inside the pen minus sea level outside [m or cm] |
| $\Delta H_i$, pair $i$ | Over-height measured at inlet $i$: LT0$i$ − LT2$i$ |
| PV, SP | Process value, setpoint |
| LIC01 | Level indicating controller of the pen (PI) |
| Framo PCC$n$ | Pump control cabinet of inlet $n$: Beckhoff CX8200 PLC and ABB ACS880 drive |
| PFC200 | WAGO 750-8212, the Heimdall PLC |
| RIO | Remote I/O: WAGO 750-362 Modbus TCP couplers (CommonRIO, InletRIO1–3) |
| OPC UA | Communication between the PLC and the Framo controllers (and the HMI) |
| HMI | Operator interface (Heimdall HMI / SiteControl) |
| VSD | Variable speed drive |
| REMOTE, HOLD, ... | Pump states published by the Framo controller (Section 3.5) |
| O-35, O-45, O-60, S-75 | Operating modes: operation (O) or service (S), with residence time in minutes |
| TBD | To be decided; see the open issues |

## 1.5 Revision note

| Rev. | Date | Change |
|---|---|---|
| A | 2026-10-21 | First issue. Based on [FD] rev. B, [FRAMO] rev. B, [PID] rev. A and the PLC branch in review (heartbeat, HOLD, per-pair PV and SafeOutput changes of 2026-10-15). Includes the review and simulator test of 2026-10-21. |
| A1 | 2026-10-21 | Reset to the PCC: held until the state change is seen, warning *Failed to reset* after 2×`ResetExpectedTime`; HOLD→REMOTE open issue reworded (the PLC follows the PCC either way); OI24 HMI Reset button. |
| B | 2026-10-22 | LIC01's output is the *total* speed of the pumps in level control, shared by the pumps in REMOTE (Section 3.2, DEC5; $K_p$ 1350 total rpm/m, one parameter set; OI5, OI6 closed). InletRIO link loss drops that inlet's pair (DEC6, OI17 closed). Level deviation alarms high/low (DEC8, OI8 in part). Automatic pump count and rotation by run time with `PumpCycleTime` (Section 3.4, DEC7, OI15 in part). Speed deviation alarm (OI9); LowPower / HighPower out of the common alarm (OI10 closed). Live test 2026-10-22 of the start, sharing, pump count and rotation (Section 5). Operator commands restructured: pen state machine with Ready and pulses only, controller modes as pulses with a status, internal commands apart (Section 3.3). Simulator tests of the sharing, the pump count, the rotation and the pen states are pending. |
| B1 | 2026-10-23 | LIC01 auto→manual bumpless: the manual output starts at the current output per pump (IN-32; before, it jumped to the last `ManualOutput`, 40 rpm if never set). |

# 2 Block diagram description

## 2.1 The process in brief

The pen PEN01 is a closed bag of about 20 000 m$^3$ (Ø 50.5 m, depth 17.3 m) with a Ø 5.5 m outlet in the bottom. Sea water is pumped in from three separate intakes at 10–30 m depth. Each intake has a fish screen (XF-3$n$), a tunnel to a Framo inlet pump (PM-3$n$, rated 20 000 m$^3$/h at 130 rpm, 60 kW rim motor) and a riser with a gooseneck over the floating collar into the pen.

The water inside the pen is kept a few centimetres higher than the sea outside. This *over-height* presses the bag outwards so that it keeps its shape, and it drives the outflow through the bottom outlet. The outlet dampers stand at a fixed opening for each operating mode; the level is therefore controlled with the *pump speed*: more speed gives more inflow, a higher level and more outflow, until inflow and outflow balance [FD] §3.1.

Before a pump can lift water over the collar, the air in the riser must be removed. Each inlet has a vacuum (air evacuation) system run by the Framo controller: a vacuum chamber TV-3$n$ on top of the riser, an ejector EJ-3$n$ driven by compressed air through valve XV3$n$A, level switches LSL3$n$/LSH3$n$ and a vent valve XV3$n$B that breaks the vacuum when the pump stops, so the pen does not drain back through the intake [FD] §3.2.1.1.

The operating modes set the flow through the pen and the level setpoint (Table 2).

Table 2: Operating modes [FD] Tabell 10, with the operating point of the design basis [DES].

| Mode | Res. time [min] | Flow [m$^3$/h] | SP $\Delta H$ [cm] | Pumps [–] | Speed $n_0$ [rpm] | Damper [%] | Comment |
|---|---|---|---|---|---|---|---|
| O-35 | 35 | 34 000 | 6.0 | 3 | 73.7 | 100 | flow above the [DES] maximum[OI12] |
| O-45 | 45 | 27 000 | 2.5 | 2 | 87.8 | 100 | |
| O-60 | 60 | 20 000 | 0.5 | 2 | 65.0 | 91 | 5 mm setpoint[OI7] |
| S-75 | 75 | 16 000 | 0.5 | 2 | 52.0 | 73 | service mode[OI7] |

With two pumps running, [FD] asks for the pumps to be alternated so that the third does not stand still for too long (HOLD in [FD])[OI15].

## 2.2 Physical units and networks

Figure 1 shows the physical units and how they are connected; Table 3 gives the role of each.

## 2.3 Simplified process and instrument diagram

Figure 2 is a simplified P&ID for one inlet ($n = 1$); inlets 2 and 3 are identical. Tags follow ISA-5.1 as on the P&ID set [PID]. Bubbles: plain = field instrument, with a bar = function in the Heimdall PLC, grey = function in the Framo controller.

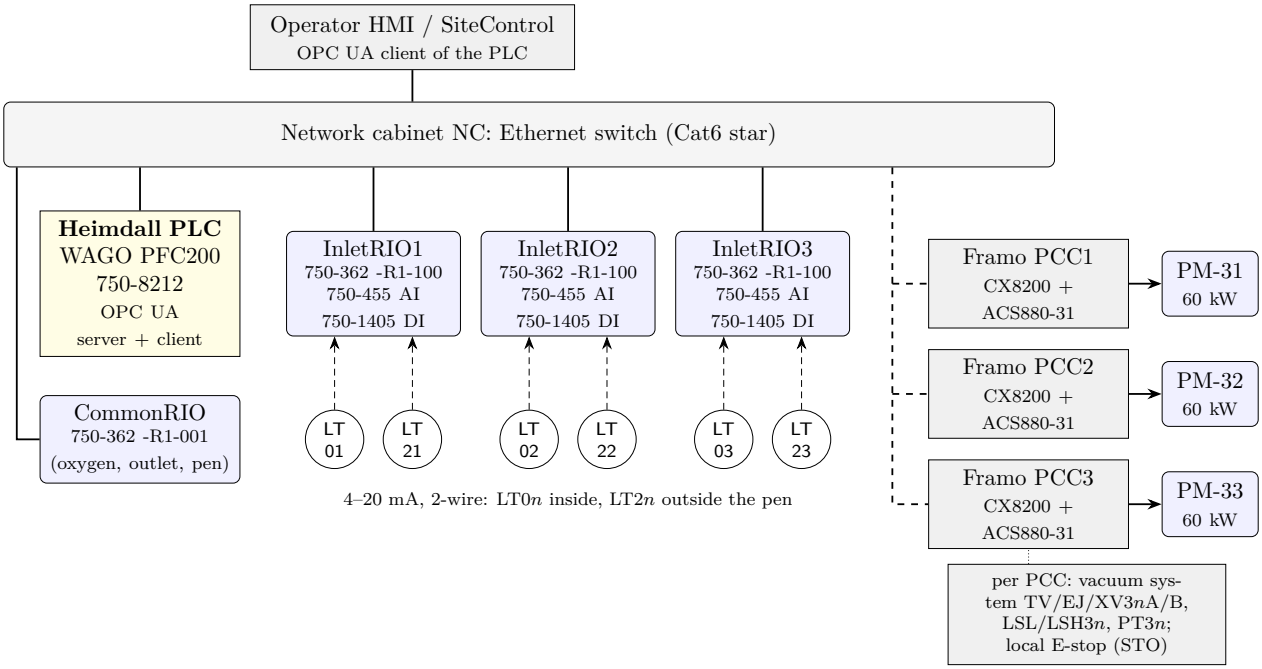



Figure 1: Physical units and networks of the inlet system (after [PID] CAS-LAY-700 and [EL]). Solid: Ethernet / Modbus TCP (PLC ↔ RIO, 100 ms); dashed: OPC UA (PLC ↔ Framo PCC); dotted arrows: 4–20 mA. IP addresses are not on any drawing[OI23].

Table 3: Physical units of the inlet system.

| Unit | Role |
|---|---|
| Framo PCC1–3 | One cabinet per inlet [EL] 874.001: Beckhoff CX8200 PLC (TwinCAT 3, OPC UA server) and ABB ACS880-31-105A-3 drive for the rim motor of PM-3$n$, local emergency stop pushbutton with Safe Torque Off (XSTO). Runs the pump's own start and stop sequences, the vacuum system and the protections, and publishes the pump state. Framo scope.[OI19] |
| Framo inlet pumps PM-31/32/33 | SR2000 rim-driven propeller pumps, 40–130 rpm, 20 000 m$^3$/h at 130 rpm, 72-pole motor (78 Hz at 130 rpm) [DES]. |
| Vacuum system (per inlet) | TV-3$n$, EJ-3$n$, XV3$n$A (air to ejector), XV3$n$B (vent), LSL3$n$, LSH3$n$, PT3$n$ (compressed air). Controlled by the PCC; the PLC only reads status.[OI11] |
| Heimdall PLC | WAGO PFC200 750-8212 in the PLC/RIO1 cabinet (UPS backed). Runs the level loop, the pen sequence and the pump supervision. No local I/O. |
| InletRIO1–3 | 750-362 Modbus TCP couplers, one per inlet. 750-455 analog inputs for LT0$n$ (word 0) and LT2$n$ (word 1); 750-1405 for cabinet power status (word 4) and bilge switches (word 5). Polled by the PLC every 100 ms. |
| CommonRIO | 750-362 coupler for the common signals (oxygen, outlet, analysis). Not used by the inlet functions. |
| Level transmitters | LT01–03 inside the pen (VEGAPULS radar on the gooseneck), LT21–23 outside (submersible pressure probe), one pair per inlet, 0–2 m, 4–20 mA 2-wire.[OI18] |
| HMI / SiteControl | Operator interface; reads and writes the PLC's variables over OPC UA. |

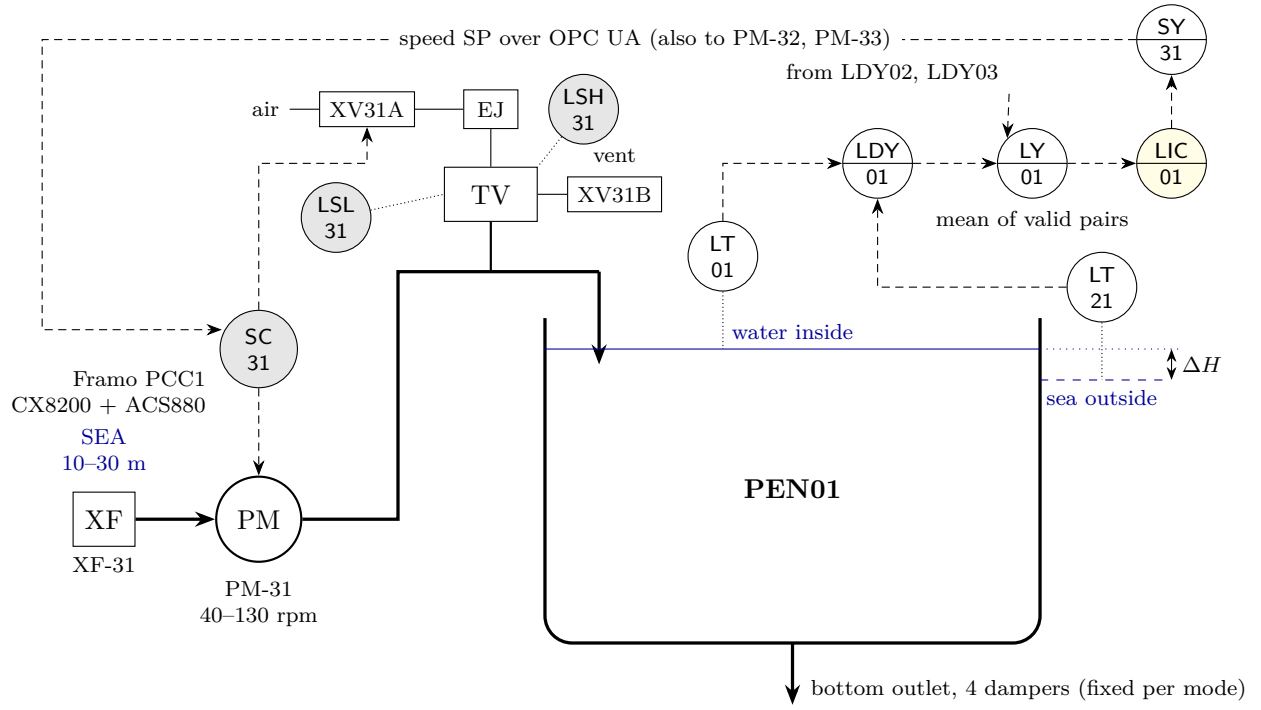



Figure 2: Simplified process and instrument diagram, inlet 1 ([PID] CAS-PID-100 and -400). LDY0$n$ forms the over-height of pair $n$, LY01 the mean of the valid pairs, LIC01 the total speed, shared as the common speed setpoint SY3$n$ of the pumps in REMOTE.

# 3 Functional diagram description

## 3.1 Top-level functions

Figure 3 shows the functions of the inlet system and the data between them. The functions F1–F3 run once per pen; F4 and F5 once per pump. The Framo controller (grey) is outside the Heimdall PLC.

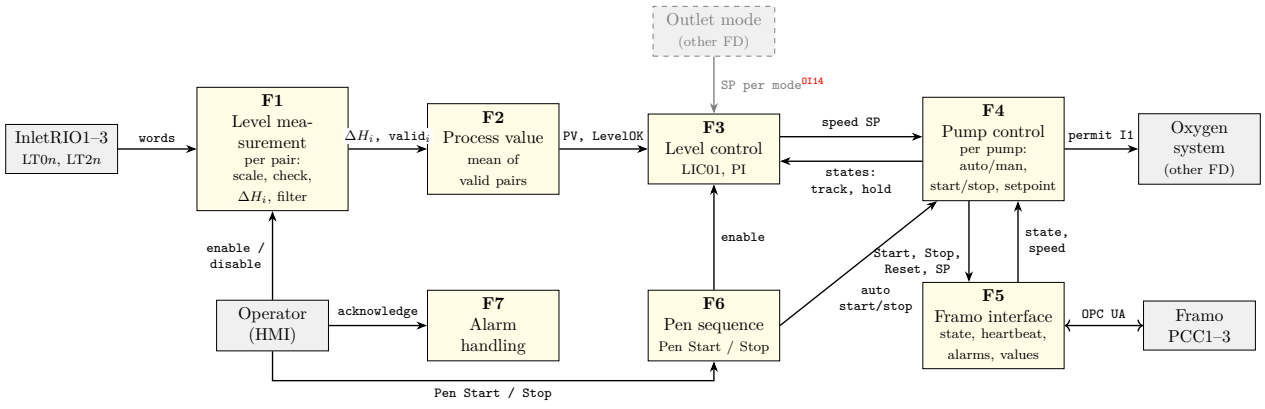



Figure 3: Top-level functional block diagram of the inlet system ([COO] Fig. 7.8 style). Not drawn: alarm conditions from F1–F5 to F7, and the operator's setpoint, LIC01 auto/manual and pump commands to F3 and F4.

Table 4: Functions and where they are implemented in [PLC].

| | Function | What it does | PLC code unit |
|---|---|---|---|
| F1 | Level measurement | Scales LT0$i$ and LT2$i$, checks their health, forms $\Delta H_i$ and filters it. | `FB.Inlet.InletSensors`, `AnalogSensor` |
| F2 | Process value | Mean of the valid $\Delta H_i$; LevelControlOK. | `FB.Pen.PenControl` |
| F3 | Level control | LIC01 (total speed) with tracking, hold, resume and safe output; share per pump in REMOTE; deviation alarms. | `InletSupervisor.LevelControl` |
| F4 | Pump control | Auto/manual, start/stop latch, speed setpoint per state, start/stop timeouts, oxygen permit. | `InletControl.InletPumpControl` |
| F5 | Framo interface | Writes commands, reads state, alarms and values; heartbeat supervision. | `PRG.FramoDataProcessing`, `InletControl.Heartbeat` |
| F6 | Pen sequence | Pen Start/Stop latch, auto start/stop of the pumps in auto, automatic pump count and rotation. | `InletSupervisor`, `.PumpCount` |
| F7 | Alarm handling | Alarm state and acknowledgement per alarm. | `AlarmManagement` instances |

### 3.2 Level control loop LIC01

**Principle.** One PI controller, LIC01, holds the over-height at the setpoint. Its output $T$ is the *total* speed of the pumps in level control (sum of rpm, DEC5). The pumps in automatic and in REMOTE share it equally, so all of them run at the same speed [FD] §3.2.2:

$$n_{\mathrm{SP}} = \mathrm{limit}_{40}^{130}\Big(\frac{T - \sum_{j \notin R} n_j}{|R|}\Big)$$

where $R$ is the set of pumps in level control that are in REMOTE and $n_j$ the actual speed of the others in level control (starting or stopping, on the PCC's own speed). Inflow goes with the total, so the loop gain does not depend on the number of pumps (one parameter set instead of [FD] Tabell 8), and a change in the pump count is made up in the same cycle instead of through the integral: a pump that trips leaves the sum and the others take over its share; a starting pump adds its speed while it ramps up and the others give way. The share (`PenInletSupervisor.AutoSpeedSetpoint`) is frozen while LIC01 is frozen (HOLD). Figure 4 is the control scheme.

**Process value.** At each inlet $i$ the PLC computes the over-height of the pair from the scaled, unfiltered readings and filters it with a 2nd-order Butterworth low-pass (cut-off $f_c = 0.05$ Hz):

$$\Delta H_i = \mathrm{LT0}i - \mathrm{LT2}i, \qquad \mathrm{PV} = \Delta H_{\mathrm{avg}} = \frac{1}{|V|}\sum_{i \in V} F\{\Delta H_i\}$$

where $V$ is the set of *valid* pairs. A pair is valid when both transmitters are enabled and healthy (750-455 status bits 1..0 = 00) and the pair itself is enabled. If either transmitter of a pair is lost, the whole pair drops out of the mean[DEC1]. [FD] asks to filter each transmitter before the difference; since the filter is linear, filtering the difference gives the same result while both are healthy, and it restarts cleanly when a pair returns [DES] §7.

**Setpoint.** The over-height setpoint in m (`Pen.Command.LevelSetpoint`, retained, default 0.06 m = O-35). [FD] wants it selected from the operating mode, with a manual override; today the operator enters it[OI14].

**Output.** Total speed, limited to $N\times$(40–130) rpm with $N$ the number of pumps in level control. The per-pump values (limits, `SafeOutput`, the operator's manual output) stay in rpm per pump and are scaled by $N$. While LIC01 is in manual its output is taken, in this order of priority, from: (1) the operator's manual output $\times N$; (2) `SafeOutput` (60 rpm) $\times N$ when no valid pair is left[DEC3]; (3) the output frozen when the heartbeat to a pump in level control was lost; (4) the summed actual speed of the pumps in level control for one cycle after the freeze (resume); (5) the summed actual speed of the starting pumps (tracking). Both transfers are bumpless: auto→manual by the operator sets the manual output to LIC01's current output per pump (total/$N$, within 40–130 rpm), so the pumps keep their speed; manual→auto continues from the manual output.

**Level deviation** (DEC8). With LIC01 in auto, alarm *Level deviation high* (*low*) when $\Delta H_{\text{avg}}$ stays more than `LevelDeviationLimit` (0.02 m) above (below) the setpoint for `LevelDeviationDelay` (5 min): the pumps cannot hold the level, at maximum speed or at minimum speed with no pump left to stop. Not raised during start-up, in manual, at `SafeOutput` or while frozen.

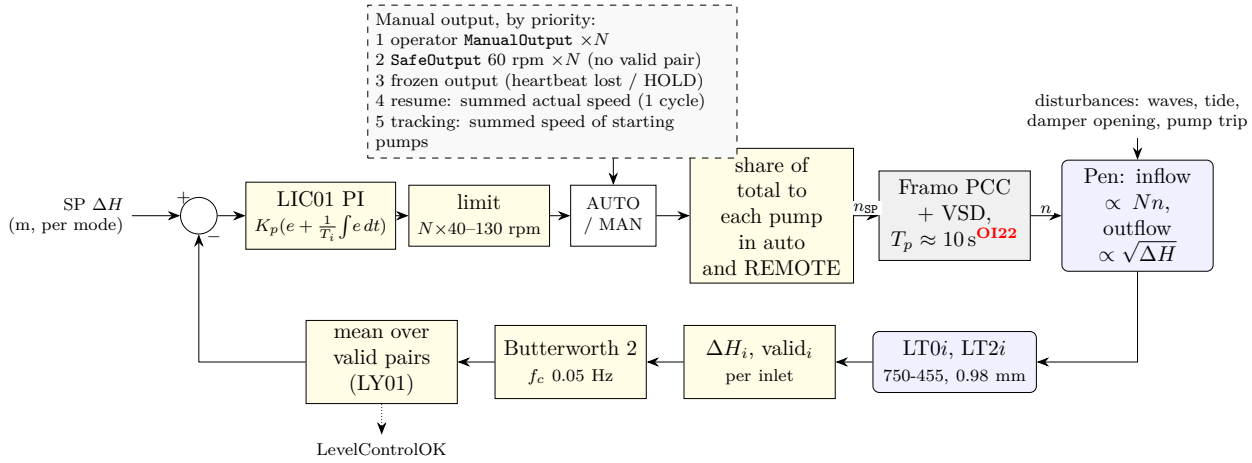



Figure 4: Control scheme of LIC01 ([PDS] SMs 1–10, [DES] §7). $N$ = number of pumps in level control.

### 3.3 Pen start and stop sequence

Figure 5 shows the sequence that the operator's Pen Start and Pen Stop commands run. Pumps in manual are left out of it; the operator starts and stops them one by one.

**Pen states** ([COO] §7.2.3; `Pen.Status.State`): *Stopped* → *Starting* on Start → *Running* when a pump in level control is in REMOTE (LIC01 takes over) → *Stopping* on Stop → *Stopped* when no pump in auto runs. Starting falls back to Stopped if no pump starts at all. Start is accepted only in Stopped and when `Pen.Status.Ready`: an enabled pump in auto is AVAILABLE (`PumpReady`) and a level pair is valid. Oxygenation is switched for the whole pen with Oxygen Start / Stop (accepted while Starting or Running; Stop turns it off).

**Command rules.** Every operator command is a pulse that the PLC clears in the same scan; one it refuses (wrong state, not ready) sets `Pen.Status.CommandRejected`. Modes are statuses: LIC01, OIC01, the pumps, the oxygen valves and the outlet go to manual and back with `Command.Manual` / `Command.Auto` pulses and show `Status.Manual`. Enable / disable of a pump or oxygen line is retained configuration (`InletConfig[i].InletPump.Enable`, `OxyConfig[i].Enable`). Commands the PLC code gives itself are in `InternalCommand` (supervisor → pump: `AutoStart` / `AutoStop`; pump → PCC: Start, Stop, Reset, speed, heartbeat), never in `Command`.

**Fresh site** (configuration set, pumps stopped; after a download pumps, lines and controllers are in auto): acknowledge the start-up alarms, check *Ready*, Pen Start, then Oxygen Start.

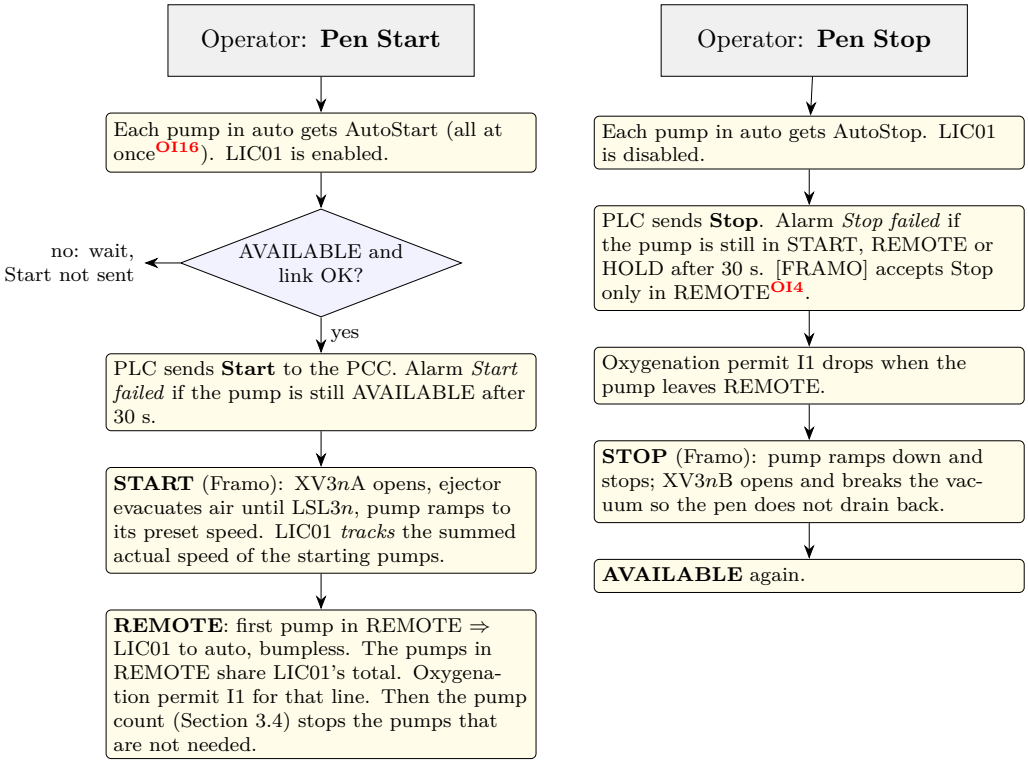



Figure 5: Pen start (left) and stop (right) sequence [FD] §3.2.1, [PLC] InletSupervisor/InletControl.

### 3.4 Pump count and rotation

While the pen runs, the PLC chooses how many pumps run and which (DEC7). The goal is to run the pumps near their operating speed rather than many pumps slowly. The PLC acts only on pumps that are *candidates*: in automatic, not UNAVAILABLE or LOCKDOWN, and enabled for it with the retained flag `InletConfig[i].InletPump.Enable`. Pumps in auto without Enable follow Pen Start / Pen Stop only; pumps in manual are left alone. The logic is in `InletSupervisor.PumpCount`.

- **Pen Start** starts every candidate (start all, then reduce). A pump that becomes a candidate while the pen runs (back to auto, reset from LOCKDOWN, enabled) is started too.
- The count logic acts only with LIC01 in auto and every pump where it is wanted (in REMOTE or AVAILABLE, none starting or stopping).
- **Stop a pump** when the others would run at or below `PumpTargetSpeed`: $T/(N-1) \leq 85$ rpm for `PumpRemoveDelay` (15 min). The pump with the most running days (`MotorRunningDays`) stops.
- **Start a pump** when the share is at or above `PumpAddSpeed`: $n_{SP} \geq 115$ rpm for `PumpAddDelay` (2 min), and a candidate is AVAILABLE. The one with the fewest running days starts. Starting is the safe direction and therefore faster than stopping.
- `PumpAddSpeed` above `PumpTargetSpeed` gives the hysteresis: 2 pumps at 115 rpm → 3 at 77 rpm; 3 at 50 rpm → 2 at 75 rpm; neither falls back at once.
- **Rotation.** When a running candidate has been in REMOTE for `PumpCycleTime` (24 h) and a candidate is AVAILABLE, the standby with the fewest running days starts first; the pump that has run its cycle stops once the standby is in REMOTE (make before break). The sharing keeps the total speed through the change. If the standby does not start, the running pump carries on.
- **Start failed.** A candidate still AVAILABLE `StartupTime` + 10 s after its start is left out of the count (alarm *Start failed*); the logic starts another one if the share stays high. It is taken back when it is no longer a candidate (operator: manual, disable) or the pen stops.

The HMI shows the number of pumps the PLC wants (`PenInletSupervisor.PumpsWanted`) and a rotation in progress (`PumpRotationActive`). [FD]'s warning for a pump idle too long and the number of pumps per operating mode are not implemented[OI15].

### 3.5 Framo pump state machine

The Framo controller owns the pump state and publishes it as one Boolean per state; the PLC turns the flags into `Inlet[i].InletPump.State` and asks for transitions with Start, Stop and Reset. Figure 6 is the state diagram of [FRAMO] rev. B; red dashed labels mark where [FD] or the PLC expect something else. Table 5 gives what the PLC does in each state.

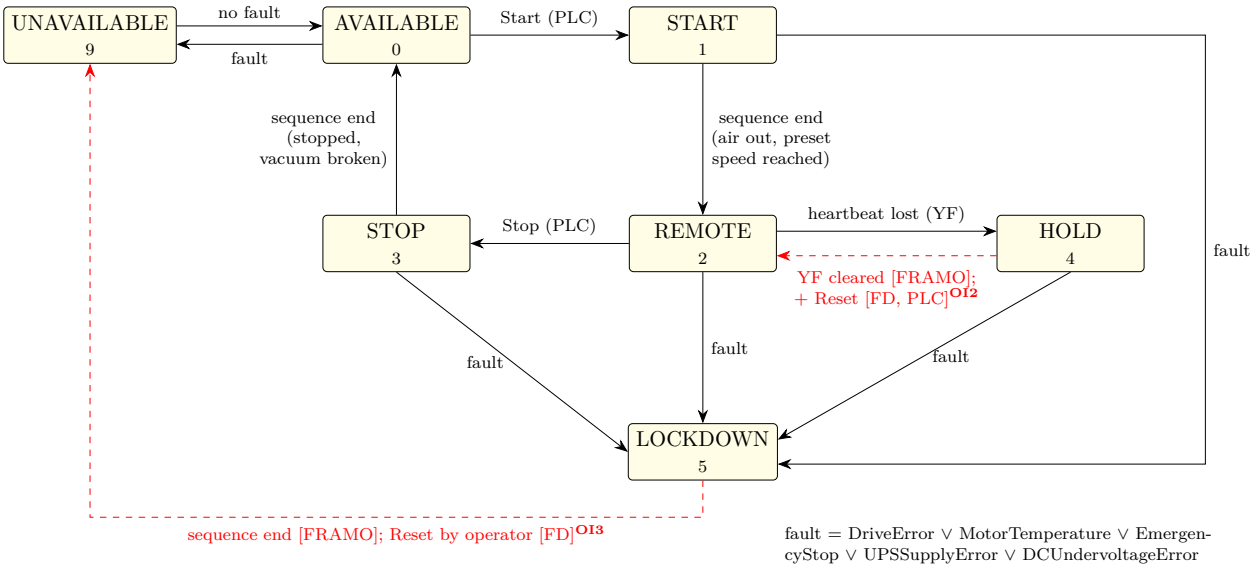



Figure 6: State transition diagram of the Framo inlet pump [FRAMO] rev. B p. 6, [COO] §7.2.3. Numbers are the values of `ENUM.InletPumpState`. In addition the PLC shows a pump UNAVAILABLE whenever it has no heartbeat answer from that PCC, whatever the PCC publishes.

Table 5: PLC action per pump state ([PLC] `InletControl.InletPumpControl`).

| State | Speed setpoint sent | Start | In LIC01 | Notes |
|---|---|---|---|---|
| UNAVAILABLE | 0 | no | no | Also shown while the link (heartbeat) is down. |
| AVAILABLE | actual speed | yes | no | Setpoint follows the actual speed so the take-over is bumpless [FD] Tabell 5. |
| START | actual speed | yes | yes | LIC01 tracks the summed speed of the starting pumps; its speed is subtracted from the total. |
| REMOTE | share of LIC01's total (auto) or operator setpoint (manual) | yes | yes | The only state in which the PLC sets the speed. Oxygenation permit I1 with a live link. |
| HOLD | actual speed | yes | no | PCC runs its preset safe speed; LIC01 frozen; oxygenation stops. |
| STOP | 0 | no | yes | |
| LOCKDOWN | 0 | no | no | Needs Reset after the fault is cleared[OI3]. |

### 3.6 Heartbeat

The PLC and each PCC supervise the link between them with a heartbeat. As implemented in the PLC (2026-10-15), the heartbeat is *one shared Boolean* `Framo_n.HeartbeatToggle` (Figure 7):

- the PLC sets it when it has been FALSE for `HeartbeatInterval` (1 s);
- the PCC clears it when it is TRUE;
- the bit always says whose turn it is, so after a (re)connect neither side waits for the other;
- only the PCC's clear (falling edge) proves that the PCC is alive. When the PCC has not cleared the bit for `HeartbeatTimeout` (10 s), the PLC raises *Communication loss*, shows the pump UNAVAILABLE, sends no Start, freezes LIC01 if the pump was in level control and removes the oxygenation permit of the line;
- the PCC, seeing no set from the PLC within its timeout, moves the pump to HOLD and runs it at a preset safe speed.

The timeout is 10 s instead of the 5 s DUT default because the round trip through the CODESYS Data Sources takes about 4.6 s in the test rig. [FRAMO] rev. B lists `HeartbeatToggle` (BOOL, RW), `HeartbeatInterval` and `HeartbeatElapsed`; [FD] describes an INT counter 0–32767. The baseline must be agreed with Framo[OI1].

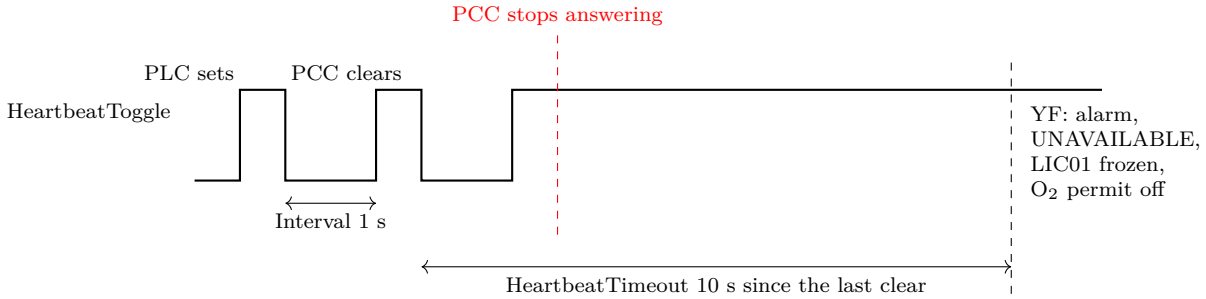



Figure 7: Heartbeat as implemented in [PLC]: one shared bit, the PLC sets it, the PCC clears it.

### 3.7 Operating modes

The operating modes (Table 2) are selected on the outlet function. A mode sets the damper opening and, per [FD], the level setpoint of LIC01 and the number of pumps (2 or 3). In [PLC] the outlet has the mode table with damper positions; the level setpoint is not taken from the mode yet[OI14]. The number of pumps follows from the flow needed (Section 3.4) instead of from the mode.

## 4 Functional requirements

### 4.1 Processes

P1 The system shall pump sea water into the pen with up to three Framo inlet pumps; normally 2 or 3 run, depending on the operating mode.

P2 The system shall hold the over-height $\Delta H_{\text{avg}}$ at the setpoint by setting one common speed for all pumps in REMOTE in level control, their share of LIC01's total speed.

P3 The system shall start and stop the pumps in automatic from one Pen Start / Pen Stop command, and let the operator start, stop and set the speed of a pump in manual.

P4 While the pen runs, the system shall run only as many pumps as the flow needs, near their operating speed, and rotate the running pumps by run time (Section 3.4).

P5 The system shall supervise the link to each Framo controller and react to its loss (HOLD).

P6 The system shall give each oxygen line its permit only while its inlet pump runs in REMOTE with a live link (interlock I1).

P7 The system shall present the pump state, the Framo values and alarms, and the levels on the HMI.

## 4.2 Inputs

Table 6: Inputs to the inlet functions (details in Tables 12 and 13).

| Input | Source | Used for |
|---|---|---|
| LT01–03, LT21–23 | InletRIO *n* words 0, 1 | PV of LIC01, display, alarms |
| Pump state flags (7 per pump) | PCC*n*, OPC UA | F4, tracking, hold, start permissive |
| MotorVelocity | PCC*n*, OPC UA | tracking, bumpless transfer, display |
| Framo alarms (11 per pump) | PCC*n*, OPC UA | alarms, info panel |
| HeartbeatToggle | PCC*n*, OPC UA | link supervision |
| Vacuum status (LSL, LSH, V001, V002, air pressure) | PCC*n*, OPC UA | display; overfill alarm[OI11] |
| Drive values (current, power, ...) | PCC*n*, OPC UA | info panel |
| 24 V field breaker XF05 status | InletRIO *n* word 4.0 | alarm (it feeds the LT loops) |
| Operator commands and parameters | HMI, OPC UA | Table 14 |

## 4.3 Outputs

- Per pump, to PCC*n* over OPC UA: Start, Stop, Reset, SpeedSetpoint [rpm], HeartbeatToggle (set).
- To the oxygen system: `Inlet[i].InletPump.Status.OxyPermit` (permit I1).
- To the HMI: states, PV, setpoint, LIC01 output and mode, alarms.

The inlet system has no hard-wired outputs; the PLC has no command path to the pumps other than OPC UA. A lost link therefore always leaves the pump to the PCC (HOLD).

## 4.4 Startup and shutdown

**PLC start (power-up, download).** Configuration and the level setpoint are retained. The pen start latch is not retained: after a PLC restart no pump is started automatically. [FD] Tabell 2–3 asks for auto-restart after a power failure, in a sequence to avoid simultaneous inrush currents; this is not implemented[OI16]. The PLC shows every pump UNAVAILABLE until its PCC has answered the heartbeat once, so no Start is sent on stale data.

**Normal start** and **normal stop**: Figure 5.

**Pump start while others run.** A pump started (in auto or manual) while another pump is already in REMOTE does not interrupt automatic control: while it starts its speed counts in LIC01's total and the pumps in REMOTE give way; it reaches REMOTE at about its share.

**Emergency stop.** Each PCC has a local E-stop pushbutton acting on the drive's Safe Torque Off; [EL] 874.001 notes that multiple E-stop actuators are connected in series. Whether the E-stops stop all three lines, and how the oxygen line is stopped on E-stop through the Heimdall PLC, is to be confirmed[OI13]. The PCC reports `EmergencyStop`; the pump goes to LOCKDOWN (or UNAVAILABLE if it was not running).

### 4.5 Fault identification and recovery

Table 7: Faults, detection, automatic response and recovery.

| Fault | Detection | Automatic response | Recovery |
|---|---|---|---|
| One level transmitter bad | 750-455 status bits ≠ 00 (wire break, over-range) for 1 s | Quality alarm; its *pair* drops out of the PV; LIC01 continues on the remaining pairs | Automatic when healthy; the filter restarts at the current value (no dip) |
| Transmitter disabled by operator | Config.Enable FALSE | Pair drops out | Operator enables it |
| All pairs lost | no valid pair | Alarm *Level measurement lost* (LAL01); LIC01 to manual at `SafeOutput` 60 rpm (interlock I6) | Back to auto bumplessly when a pair is valid |
| InletRIO $n$ link lost | Modbus TCP `CommOK` FALSE | Alarm *InletRIO comm. error*; both transmitters of inlet $n$ count as bad, so its pair drops out of the PV (DEC6) | Automatic |
| 24 V field breaker XF05 tripped | InletRIO word 4.0 | Alarm; both transmitters of that inlet read bad ⇒ pair drops out | Reset XF05 (HMI, DO 0.0) |
| Framo link / heartbeat lost | no clear for 10 s | Alarm *Communication loss*; PLC shows UNAVAILABLE; LIC01 output frozen if the pump was in level control; oxygenation permit off. PCC: HOLD at preset speed | Link back: PCC still in HOLD ⇒ operator Reset[OI2]; when REMOTE, LIC01 resumes from the actual speed |
| Pump trip / fault | PCC alarm flags; state LOCKDOWN | Common alarm *Pump $n$ failed*; pump leaves level control; the remaining pumps take over its share in the same cycle (DEC5); if their share stays ≥ `PumpAddSpeed`, a standby candidate is started; permit off | Fix the fault, operator Reset[OI3]; as a candidate it is started again while the pen runs |
| Start not acknowledged | Start sent, still AVAILABLE after 30 s | Alarm *Start failed*; a candidate is left out of the pump count and another one is started if the share stays high (Section 3.4) | Operator |
| Stop not acknowledged | Stop sent, still in START, REMOTE or HOLD after 30 s | Alarm *Stop failed* | Operator, locally at the PCC |

| Fault | Detection | Automatic response | Recovery |
|---|---|---|---|
| Reset not answered | Reset given in HOLD or LOCKDOWN, pump still in that state after 2×`ResetExpectedTime` (10 s) | Reset withdrawn; warning *Failed to reset* (no acknowledge) | Check the PCC; give Reset again |
| Speed deviation | $\lvert n_{SP} - n \rvert >$ `SpeedDeviationLimit` (10 rpm) for `SpeedDeviationDelay` (30 s), in REMOTE | Alarm *Speed deviation* | Operator; automatic when the pump follows again |
| Vacuum chamber overfill | LSH3$n$ | not implemented (5 s delay in [FD])[OI11] | – |
| Emergency stop | PCC `EmergencyStop` | Pump stops (STO); alarm; LOCKDOWN | Release E-stop, Reset[OI13] |
| LIC01 at its limit | `LimitsActive` | Alarm; integral held (anti-windup of `Util.PID`) | Automatic |
| Level not held | $\lvert \Delta H_{avg} - SP \rvert >$ the limit for the delay, LIC01 in auto | Alarm *Level deviation high / low* | Operator; automatic when back within the limit |

## 4.6 Interlocks

| ID | Interlock | Source | In PLC |
|---|---|---|---|
| I1 | Oxygenation permit line $n$ = PM-3$n$ in REMOTE AND heartbeat OK. | [FD] §3.2.1, [PID] -100 | yes |
| I2 | 2oo3 high-high LS7$n$mHH in inlet box $n$ stops PM-3$n$. | [FD] §3.2.6, [PID] -600 | no[OI20] |
| I3 | Group start only with the outlet open. | PLC function flow diagram | no[OI20] |
| I6 | No valid level pair: LIC01 to manual at SafeOutput. | [PID] -400 | yes |
| – | Start only when the pump is AVAILABLE and the link is OK. | [FD] Tabell 5 | yes |
| – | Emergency stop: Framo E-stop stops the pump (STO); oxygen line to stop. | [FD] Tabell 2, [EL] | partly[OI13] |

## 4.7 Communication

- **PLC ↔ Framo PCC$n$:** OPC UA. The PLC is the client (CODESYS Data Sources `Framo_1. .3`); each PCC is a server (TwinCAT TF6100). Signal list: Table 13. Start, Stop and Reset are events acted on at the rising edge[OI4]. The PLC holds Reset TRUE from the operator's command until it reads the pump out of HOLD or LOCKDOWN, at most 2×`ResetExpectedTime`; then it withdraws Reset and warns *Failed to reset*.
- **PLC ↔ InletRIO $n$:** Modbus TCP, function code 23, every 100 ms.
- **HMI ↔ PLC:** OPC UA (PLC is the server).

# 5 Non-functional requirements

## 5.1 Control law and tuning

LIC01 is a PI controller (CODESYS `Util.PID` wrapped in `PIDController`):

$$T(t) = K_p\Big(e(t) + \frac{1}{T_i}\int e\,dt\Big) + Y_{\text{offset}}, \qquad e = \text{SP} - \Delta H_{\text{avg}},$$

the total speed of the pumps in level control, limited to $N\times$(40–130) rpm and shared as in Section 3.2; the integral is held while the output is at a limit. Manual→auto resets the PID with $Y_{\text{offset}} = Y_{\text{prev}} - K_p e$ (bumpless). Auto→manual (`LevelController.Status.Manual` rising) sets `LevelController.Command.ManualOutput` $= Y_{\text{prev}}/N$ (bumpless).

Table 8: LIC01 start values.

| Parameter | [PLC] | [FD] Tabell 8 | Comment |
|---|---|---|---|
| $K_p$ | 1350 total rpm/m | P = 0.5 / 1 / 2 for 1 / 2 / 3 pumps | = the former 450 rpm/m per pump with 3 pumps (≈ [FD] P = 1: 90 rpm over 0.20 m). With the total output one set holds for every pump count (DEC5) |
| $T_i$ | 60 s | 600 s | 600 s gives a closed loop of about 16 min [DES]; 60 s tested[DEC4] |
| $T_d$ | 0 | – | PI |
| Output | $N\times$40–130 rpm (total) | 40–130 rpm | per pump 40–130 rpm |
| PV scaling | m | 0–20 cm | |
| Margins (worst case O-35, 3 pumps) | PM 76°, GM 13.2 dB, $1/\omega_c$ = 100 s | – | [DES] §7, dead time $2.5\theta$ |

**Simulator tests** (PFC200 against CasRioSim/CasFramoSim, O-35, 3 pumps, $K_p$ 450 per pump, $T_i$ 60 s, before the sharing):

- start-up: pumps reach REMOTE at about 87 rpm, take-over bumpless;
- setpoint step +2 cm: settles in about 100 s, no overshoot;
- one of three pumps lost: the level dips 3.4 cm; the pump rejoining gives 2.7 cm overshoot. There is no feed-forward on the number of pumps, and the integral winds up against 130 rpm while two pumps carry the load.

**Baseline** (2026-10-22, same rig): pump 3 out ⇒ level 6.0 → 3.0 cm, about 4 min back; pump 3 restart ⇒ 9.1 cm, mostly during its START (its own preset speed while the others still ran fast). **Change** (DEC5): the output is the total speed, shared by the pumps in REMOTE, so a trip or a start is compensated in the same cycle. An I-PD form would avoid the setpoint kick on mode changes.

**Live test** (2026-10-22, PFC200 10.1.2.41 on `main` after Reset Origin, CAS plant simulator 10.1.2.150, O-35):

- Pen Start from a drained pen (−3.4 cm): Starting → Running when the pumps reached REMOTE; LIC01 tracked the summed speed (260 rpm) and handed over at 85.5 rpm per pump (bumpless); level peak 6.6 cm (0.6 cm overshoot), settled at 6.1 cm in about 2.5 min with three pumps at 74 rpm.

- Pump stopped by the pump count (test values `PumpTargetSpeed` 115 rpm, delay 60 s): the pump with the most running days stopped, the other two took its share in the same cycle; level dip 6.0 → 5.2 cm, back to 6.3 cm; two pumps at about 112 rpm, no flapping.
- Rotation (test value `PumpCycleTime` 2 min): the standby started first and the old pump stopped once the new one was in REMOTE; level 5.1–7.1 cm during each handover (register IN-31). With a short cycle the rotations follow each other, as every running pump has passed it.
- Commands: $O_2$ Start with the pen Stopped was refused (`CommandRejected`); all pulses were cleared by the PLC in the same scan.
- Not yet tested: pump trip, level and speed deviation alarms, RIO link loss with the new code.

**Small setpoints.** O-60 and S-75 use a 0.5 cm setpoint, against about 1 mm resolution of the over-height, ±6 mm combined transmitter uncertainty and waves up to $H_s = 2$ m. Whether this is controllable, and with what filter, is open[OI7].

## 5.2 Cycle times and tasks

Table 9: Tasks and timing ([PLC] Task Configuration of PFC200_Sim).

| Task | Cycle | Inlet content |
|---|---|---|
| `Task` | 20 ms | `Main`: InletSupervisor (LIC01, sequence), InletControl[0..2] (pump control, heartbeat, InletSensors), PenControl (PV); `IO_Processing` (InletRIO masters) |
| `OPC_Processing` | 50 ms | `FramoDataProcessing`: maps Framo_$n$ to/from GVLMain |
| `DatasourcesTask` | 50 ms | OPC UA client (Data Sources) to the PCCs |
| Modbus TCP poll | 100 ms | InletRIO $n$, FC23 |

A level reading reaches LIC01 two `Task` cycles after it is read; the speed setpoint reaches the PCC within some tenths of a second plus the OPC UA publishing time. This is small against the level loop's dead time of about 2 s and time constant of 28–40 s [DES]. Task watchdogs are disabled in the test application[OI23].

## 5.3 Signal resolution and filtering

- **750-455:** value in bits 14..4 (0..2047), bits 3..2 reserved, bits 1..0 diagnostics; 20 mA = `0x7FF0`[DEC2]. On the 0–2 m span one step is 0.98 mm; the over-height, the difference of two readings, moves in steps of about 1 mm. The PLC scales Raw · (Max − Min)/32760 + Min (20 mA reads 99.98 % of span).
- **Transmitter accuracy:** inside ±2 mm (radar), outside ±4 mm (pressure, 0.2 % of an assumed 2 m full scale)[OI18].
- **Filter:** 2nd-order Butterworth per pair, $f_c = 0.05$ Hz, bilinear with pre-warping, sample time = task cycle read from the runtime, LREAL coefficients. A 1 cm step reaches 63 % in 5.6 s with 4 % overshoot; 4–10 s waves are attenuated by about 19 dB [DES]. The filter restarts at the current value when a pair becomes valid.
- The per-transmitter filters (`FilterFc` 0.05 Hz) are for display only.
- **Signal direction:** raw 0 (4 mA) is assumed to be the lowest level; to be read in the transmitters. Inside and outside must agree, or the sign of LIC01 is wrong[OI18].

## 5.4 Alarm functions

Alarm states are handled by `AlarmManagement` (raised on the condition, acknowledged on the rising edge of the acknowledge bit). [AL] defines no priorities or acknowledgement classes; the column Class is a proposal (A = alarm, operator action; W = warning; N = notification only)[OI21].

Table 10: Alarms of the inlet system ([FD] Tabell 7 and 9, [AL], [PLC]). Per pump $n$ = 1..3 unless noted.

| Alarm | Condition | Class | Action | PLC |
|---|---|---|---|---|
| Communication loss pump $n$ | PCC has not cleared HeartbeatToggle for 10 s | A | UNAVAILABLE; LIC01 frozen; $O_2$ line $n$ stops | yes |
| Start of pump $n$ failed | Start sent and still AVAILABLE after StartupTime (30 s) | A | none | yes |
| Stop of pump $n$ failed | Stop sent and still in START, REMOTE or HOLD after ShutdownTime (30 s) | A | none | yes |
| Failed to reset pump $n$ | Reset given in HOLD/LOCKDOWN, no state change within 2×ResetExpectedTime (10 s); clears when the pump leaves HOLD/LOCKDOWN or on a new Reset | W (auto-ack) | Reset withdrawn | yes |
| Pump $n$ failed (common) | any of DriveError, DriveCommunication, DriveNotInRemote, MotorTemperature, EmergencyStop, DCUndervoltageError, PumpStopped, UPSSupplyError, EmergencyRunMode | A | shown on info panel | yes[OI10] |
| Individual Framo alarms | each flag above, own alarm | A | logged | yes |
| Pump $n$ low / high power | LowPower / HighPower from PCC; not in the common alarm | N (auto-ack) | none, notification only [FD] | yes |
| Speed deviation pump $n$ | $\lvert n_{SP} - n \rvert >$ `SpeedDeviationLimit` (10 rpm) for `SpeedDeviationDelay` (30 s), in REMOTE (`Inlet[i].InletPump.Alarm.SpeedDeviation`) | W | none | yes |
| Vacuum overfill line $n$ | LSH3$n$ for 5 s (configurable) | A | none | no[OI11] |
| Emergency stop activated | EmergencyStop from PCC (hard-wired TBD) | A | pump stops | via Framo[OI13] |

| Alarm | Condition | Class | Action | PLC |
|---|---|---|---|---|
| Level deviation high / low | $\Delta H_{\text{avg}}$ more than `LevelDeviationLimit` (0.02 m) above / below SP for `LevelDeviationDelay` (5 min), LIC01 in auto (`Pen.Alarm.LevelDeviationHigh`, `.LevelDeviationLow`) | A | none | yes |
| Low level in pen | $\Delta H_{\text{avg}} <$ TBD for TBD s | A | none | no[OI8] |
| High level in pen | $\Delta H_{\text{avg}} >$ TBD for TBD s | A | none | no[OI8] |
| Level transmitter fault (LT01–03, LT21–23) | 750-455 status bits $\neq$ 00 for 1 s | W | pair drops out of PV | generic quality alarm[OI8] |
| Level measurement lost (LAL01) | no valid pair (pen) | A | LIC01 at SafeOutput | yes |
| LIC01 output at limit / overflow | `LimitsActive`, `Overflow` (pen) | W | none | yes |
| InletRIO $n$ communication error | Modbus `CommOK` FALSE | A | pair $n$ drops out of the PV | yes |
| InletRIO $n$ field power | XF05 tripped | A | LTs of inlet $n$ lost | yes |

## 5.5 Parameters

All parameters are written from the HMI and retained: `Main.UpdateRetain` keeps them in step with `GVLRetain`. Defaults from `ConfigValues` apply on a fresh download (rev B added fields to `PenConfigDUT`: download with Reset Origin).

Table 11: Configurable parameters of the inlet system.

| Parameter | Default | Unit | Meaning |
|---|---|---|---|
| `PenConfig.LevelController.Kp` | 1350 | rpm/m | LIC01 gain on the total speed |
| `PenConfig.LevelController.Ti` | 60 | s | integral time (new value takes effect at the next start or manual→auto) |
| `PenConfig.LevelController.Td` | 0 | s | derivative time |
| `PenConfig.LevelController.MinOutput` / `MaxOutput` | 40 / 130 | rpm | speed range per pump (LIC01: $\times N$) |
| `PenConfig.LevelController.SafeOutput` | 60 | rpm | speed per pump with no valid level pair |
| `PenConfig.LevelDeviationLimit` / `LevelDeviationDelay` | 0.02 / 5 | m / min | level deviation alarms; limit 0 = off |
| `PenConfig.PumpTargetSpeed` | 85 | rpm | stop a pump when the others would run at or below this; 0 = never |
| `PenConfig.PumpAddSpeed` | 115 | rpm | start a pump when the share is at or above this; above `PumpTargetSpeed`; 0 = never |
| `PenConfig.PumpRemoveDelay` / `PumpAddDelay` | 15 / 2 | min | how long the stop / start condition must hold |
| `PenConfig.PumpCycleTime` | 24 | h | time in REMOTE before a standby takes over; 0 = no rotation |

| Parameter | Default | Unit | Meaning |
|---|---|---|---|
| `InletConfig[i].InletPump.Enable` | TRUE | | pump $i$ takes part in the pump count and rotation |
| `Pen.Command.LevelSetpoint` | 0.06 | m | over-height setpoint (O-35) |
| `LevelInsideConfig[i]`: Enable, MinValue, MaxValue | TRUE, 0, 2.0 | m | LT0$i$ in use and range |
| `LevelInsideConfig[i]`: FilterFc, QualityEnabled | 0.05, TRUE | Hz | display filter; quality alarm |
| `LevelOutsideConfig[i]`: as inside | as inside | | LT2$i$ |
| `OverHeightConfig[i]`: Enable, FilterFc | TRUE, 0.05 | Hz | pair in use; cut-off of the pair's filter (0 = off) |
| `InletConfig[i].HeartbeatInterval` | 1 | s | PLC sets the bit this long after the PCC cleared it |
| `InletConfig[i].HeartbeatTimeout` | 10 (DUT 5) | s | link-loss detection |
| `InletConfig[i].StartupTime` / `ShutdownTime` | 30 / 30 | s | start / stop failed alarms |
| `InletConfig[i].SpeedDeviationLimit` / `SpeedDeviationDelay` | 10 / 30 | rpm / s | speed deviation alarm in REMOTE; limit 0 = off |
| `InletConfig[i].ResetExpectedTime` | 5 | s | expected PCC response to Reset (OPC UA round trip ≈4.6 s); warning *Failed to reset* after 2× |
| `InletRIOConfig[i]`: IP, Port | sim: 10.1.2.150, 5121–5123 | | coupler address (site plan TBD) |
| Data sources `Framo_1..3` | per project | | OPC UA endpoint of each PCC OI8 OI9 OI11 OI14 |
| Not yet defined: L/H level limits and delays, speed deviation limit, vacuum overfill delay (5 s), mode table SP | TBD | | |

## 5.6 Interfaces: instruments and actuators

Table 12: Instruments, actuators and PLC functions of the inlet system ([PID], [EL], [DES]). Use: C = control, M = monitor, A = alarm.

| Tag | Function | Location | Signal / interface | Range | Resolution | Use |
|---|---|---|---|---|---|---|
| LT01–03 | Level inside the pen, at inlet $n$; VEGAPULS radar OI18 | Gooseneck, inlet $n$ | 4–20 mA 2-wire, InletRIO $n$ -R1-106 ch1, word 0 | 0–2 m | 0.98 mm; ±2 mm | C, M, A |
| LT21–23 | Sea level outside the pen, at inlet $n$; submersible pressure probe | Outside the collar, inlet $n$ | 4–20 mA 2-wire, InletRIO $n$ ch2, word 1 | 0–2 m (assumed) | 0.98 mm; ±4 mm | C, M, A |
| LDY01–03 | Over-height of pair $n$, filtered | PLC | computed | 0–0.20 m | ≈1 mm | C |
| LY01 | Mean over-height of valid pairs (PV) | PLC | computed | 0–0.20 m | – | C, A |

| Tag | Function | Location | Signal / interface | Range | Resolution | Use |
|---|---|---|---|---|---|---|
| LIC01 | Level controller, PI | PLC | output to SY31–33 | 40–130 rpm | – | C |
| LAL01 | Level measurement lost | PLC | alarm | – | – | A |
| PM-31–33 | Framo SR2000 inlet pump, rim motor 60 kW | Intake $n$ | via PCC$n$ (ACS880) | 40–130 rpm, 20 000 $m^3/h$ | – | actuator |
| SC31–33 | Framo pump controller CX8200 + ACS880-31-105A-3 | PCC$n$ | OPC UA server | – | – | – |
| SY31–33 | Speed setpoint to pump $n$ | PLC → PCC$n$ | OPC UA SpeedSetpoint | 40–130 rpm | REAL | C |
| SI31–33 | Actual pump speed | PCC$n$ | OPC UA MotorVelocity | 0–130 rpm | REAL | C, M |
| HS31–33 | Start / Stop / Reset pump $n$ | PLC → PCC$n$ | OPC UA events | – | – | C |
| YI31–33 | Pump state | PCC$n$ | OPC UA 7 flags | 7 states | – | C, M |
| UA31–33 | Pump alarms (11 flags) | PCC$n$ | OPC UA alarms | – | – | A |
| KY31–33 | Heartbeat | PLC ↔ PCC$n$ | OPC UA BOOL | – | – | C, A |
| LSL31–33 | Vacuum chamber low level (air evacuated) | TV-3$n$ | OPC UA (Framo LSL01) | on/off | – | M |
| LSH31–33 | Vacuum chamber high level (overfill) | TV-3$n$ | OPC UA (Framo LSH02) | on/off | – | A[OI11] |
| PT31–33 | Compressed air pressure to ejector | Air line, inlet $n$ | OPC UA analog | Framo range | REAL | M |
| XV31A–33A | Air valve to ejector EJ-3$n$ | Inlet $n$ | status OPC UA (V001?) | open/closed | – | M[OI11] |
| XV31B–33B | Vacuum break (vent) valve | TV-3$n$ | status OPC UA (V002?) | open/closed | – | M[OI11] |
| E-stop PCC$n$ | Local emergency stop, STO | PCC$n$ door | OPC UA EmergencyStop | on/off | – | A[OI13] |
| YA30 (n) | 24 V field breaker XF05 tripped (LT supply) | RIO $n$ cabinet | InletRIO $n$ DI word 4.0 | on/off | – | A |

## 5.7 Interfaces: OPC UA signals to and from the Framo controllers

Per pump $n$ the PLC uses the Data Source `Framo_n`; the GVL side is `GVLMain.Inlet[n-1].InletPump`. Direction is seen from the PLC (R = read, W = written).

Table 13: OPC UA signal list, PLC ↔ Framo PCC*n* ([FRAMO] rev. B, [FD] Tabell 6, [PLC] FramoDataProcessing).

| Signal (Data Source name) | Type | Dir. | Use in PLC | Remark |
|---|---|---|---|---|
| MotorVelocity (+ ValueRange Low/High) | AnalogValue | R | tracking, bumpless transfer, speed indication | range written by Framo? OI19 |
| MotorFrequency, MotorCurrent, MotorPower, MotorVoltage, MotorTorque, PowerConsumptionCounter, DCLinkVoltage, VSDHeatSinkTemperature, VSDOperatingDays | AnalogValue | R | info panel | [FRAMO]: PowerConsumtionCounter |
| MotorRunningDays | AnalogValue | R | info panel; run-time order of pumps (`SortSequence`) for the pump count and rotation | |
| DriveError, DriveCommunication, DriveNotInRemote, MotorTemperature, EmergencyStop, DCUnderVoltageError, PumpStopped, UPSSupplyError, EmergencyRunMode | Alarm | R | own alarm each; common alarm | |
| LowPower, HighPower | Alarm | R | notification (auto-ack), not in the common alarm | |
| PumpStateUnavailable, ... Available, Start, Stop, Remote, Hold, Lockdown | Event | R | `InletPump.State` (first TRUE flag) | |
| Start, Stop, Reset | Event | W | commands, rising edge | OI3 OI4 |
| SpeedSetpoint | AnalogValue | W | LIC01 output in REMOTE, else actual speed or 0 | [FRAMO], [FD]: Velocity-Setpoint OI19 |
| HeartbeatToggle | BOOL | R/W | PLC sets, PCC clears | OI1 |
| HeartbeatInterval, HeartbeatElapsed | TIME | – | not mapped | OI1 |
| PressureCompressedAir, LevelSwitchLSL01, LevelSwitchLSH02, StatusV001, StatusV002 | Analog / BOOL | R | vacuum system status | not in [FRAMO] / [FIL] OI11 |

Table 14: Operator functions for the inlet system (PLC variables in [PDS] Table “Commands”).

| Function | PLC variable | Note |
|---|---|---|
| Pen Start / Pen Stop | `Pen.Command.Start`, `.Stop`; shows `Pen.Status.State`, `.Ready`, `.CommandRejected` | pulses; Start only when Ready; starts the enabled pumps in auto |
| Oxygen Start / Stop | `Pen.Command.OxygenStart`, `.OxygenStop`; shows `Pen.Status.OxygenOn` | pulses; enabled lines follow their inlet pumps |
| Pump count and rotation | `InletConfig[i].InletPump.Enable`; shows `PenInletSupervisor.PumpsWanted`, `.PumpRotationActive` | take a pump out of the count without manual |
| Level setpoint | `Pen.Command.LevelSetpoint` | m; from the mode in future[OI14] |
| LIC01 auto / manual, manual output | `LevelController.Command.Manual`, `.Auto` (pulses; mode in `.Status.Manual`), `.ManualOutput` | rpm per pump |
| Pump auto / manual | `Inlet[i].InletPump.Command.Auto`, `.Manual` (pulses; mode in `.Status.Manual`) | default auto |
| Pump start / stop / speed in manual | `Inlet[i].InletPump.Command.Start`, `.Stop`, `.Setpoint` | speed used in REMOTE |
| Reset pump (HOLD, LOCKDOWN) | `Inlet[i].InletPump.Command.Reset` | pulse, accepted only in HOLD or LOCKDOWN; the PLC clears it and holds the PCC’s Reset until it sees the state change. HMI button per pump[OI24] |
| Enable/disable transmitter or pair | `...LevelInsidePen.Config.Enable`, `...PenOverHeight.Config.Enable` | takes a pair out of the PV |
| Info panel per pump | `Inlet[i].InletPump.Input.*`, `.State` | [FD] Figur 3 |
| Alarm list and acknowledgement | `*.Alarm.*`, `*.AlarmAck.*` | classes TBD[OI21] |
| Parameters (Table 11) | `GVLMain.PenConfig`, `InletConfig`, ... | access level TBD |

### 5.8 HMI operator functions

## 6 Open issues and decisions

### 6.1 Open issues

Each item names the sources that disagree and, where there is one, a proposal. “Framo” items need agreement with the supplier.

**OI1 Heartbeat baseline.** PLC: one shared BOOL, PLC sets / PCC clears, neither side waits after connect, interval 1 s, timeout 10 s (DUT default 5 s). [FRAMO] rev. B: HeartbeatToggle with HeartbeatInterval and HeartbeatElapsed. [FD]: INT counter 0–32767. Agree one with Framo, incl. the PCC’s own timeout and whether the PLC writes Interval.

**OI2 HOLD → REMOTE.** [FD] requires an operator Reset; [FRAMO] rev. B returns to REMOTE

automatically when the heartbeat is back. The PLC works either way: it follows the state the PCC reports (LIC01 frozen while in HOLD, bumpless resume in REMOTE) and sends Reset only when the operator gives it. Open with Framo: which behaviour the PCC implements, and whether the PCC's own panel also has a local Reset.

**OI3** **LOCKDOWN exit and Reset.** [FD]: operator Reset after the fault is fixed. [FRAMO]: LOCKDOWN ends with "sequence end" to UNAVAILABLE, no Reset. What Reset does (all alarms? edge?) is not described (Framo review comment).

**OI4** **Stop and command semantics.** [FRAMO] rev. B accepts Stop only in REMOTE (not in START or HOLD). The PLC sends Stop in any state and raises Stop failed for START, REMOTE and HOLD. Are Start/Stop/Reset edge- or level-triggered?

**OI5** **LIC01 parameter sets.** *Closed in rev B* by DEC5: with the total output the loop gain does not depend on the pump count; one set ($K_p$ 1350 total rpm/m, $T_i$ 60 s). [FD] Tabell 8 to be updated.

**OI6** **Pump-count transients.** *Closed in rev B* by DEC5 (sharing of the total); to be confirmed in the simulator. I-PD against setpoint kick still to consider.

**OI7** **0.5 cm setpoint (O-60, S-75).** Controllable with ≈1 mm resolution, ±6 mm uncertainty and waves? Decide the setpoint, filter and acceptance criterion.

**OI8** **Pen level alarms.** Deviation alarms high / low implemented (DEC8; 0.02 m, 5 min, to be confirmed). Low and high level alarms of [FD] Tabell 9 / [AL] are not in the PLC; limits and delays TBD. The per-transmitter fault exists only as the generic quality alarm; map it to the [AL] text per tag.

**OI9** **Speed deviation alarm** (in REMOTE) of [FD] Tabell 7: implemented in rev B (10 rpm, 30 s, to be confirmed against Framo's speed response).

**OI10** **LowPower / HighPower.** *Closed in rev B*: removed from `PumpCommonError`; their own alarms are notifications without acknowledge, as [FD].

**OI11** **Vacuum system.** Alarms not implemented (`VacuumSystemAlarmDUT` is empty; LSH overfill with 5 s delay). The vacuum signals are not in [FRAMO] or [FIL]. Tag mapping to confirm: Framo LSL01/LSH02/V001/V002 vs [PID] LSL3$n$/LSH3$n$/XV3$n$A/B vs [FD] LSL31/LSH32/XV31/XV32.

**OI12** **Flow figures.** [FD] Tabell 10 O-35 = 34 000 m$^3$/h; [DES] maximum 27 000 m$^3$/h; 3 pumps × 20 000 m$^3$/h rated. Framo comment: "26 000 ok for emergency mode?". Align.

**OI13** **Emergency stop.** Do the Framo E-stops (in series, [EL] 874.001) stop all three lines? How is oxygenation stopped on E-stop through the Heimdall PLC (no E-stop input wired today)? [FD] alarm "Nødstopp aktivert" is HOLD.

**OI14** **Setpoint from operating mode.** [FD]: LIC01 setpoint chosen from the mode, with manual override. PLC: operator-entered setpoint only; the outlet mode table has damper positions only.

**OI15** **Number of pumps and rotation.** Implemented in rev B (DEC7, Section 3.4): stop and start by the share against `PumpTargetSpeed` / `PumpAddSpeed`, rotation by run time after `PumpCycleTime`, next pump when a start fails. Still open: [FD]'s warning when a pump has been idle too long; thresholds, delays and cycle time to be confirmed with Framo (start and stop wear, minimum run time); the level swing of about ±1 cm during a rotation handover (live test 2026-10-22, register IN-31); `NumberOfInletsToRun` unused.

**OI16** **Auto-restart and start order.** [FD] Tabell 2–3: restart after power failure in a sequence (2a/2b/2c). PLC: pen start latch not retained; Pen Start sends Start to all pumps at once.

**OI17** **InletRIO link loss.** *Closed in rev B* by DEC6: the pair is invalid while its RIO has no communication (tested in the simulator 2026-10-22).

**OI18** **Level transmitters.** Signal direction (4 mA = lowest level?) to be read in the sensors; inside model VEGAPULS C21 ([DES]) vs C22 ([PID] loop sheet); outside type and 2 m full scale assumed.

**OI19** **Framo interface details.** Names differ (SpeedSetpoint vs VelocitySetpoint, DCUnderVoltageError, PowerConsumptionCounter); is the MotorVelocity range written by Framo? Hardware CX8190 ([FRAMO]) vs CX8200 ([EL]); pump SX1000 ([EL]) vs SR2000 ([FD]).

**OI20** **Interlocks from other systems.** 2oo3 LS7$n$mHH trip of PM-3$n$ (I2) and "group start only with the outlet open" (I3) are not in the PLC.

**OI21** **Alarm classes.** [AL] has no priority or acknowledgement class; define them.

**OI22 Framo dynamics and preset speeds.** Speed response $T_p$ (10 s assumed), START preset speed (88 rpm assumed), HOLD safe speed and ramp rates: to be given by Framo.

**OI23 Platform.** Task watchdogs disabled in the test application; IP plan and network drawing missing ([PID] C3).

**OI24 Operator Reset on the HMI.** A Reset button per inlet pump (enabled in HOLD and LOCKDOWN, with confirmation) is in review for the test HMI (heimdall-cas-hmi); the production HMI needs the same function.

## 6.2 Decisions

**DEC1** 2026-10-15 (PLC, in review): PV = mean of the per-pair filtered $\Delta H_i$ over the valid pairs; a lost transmitter drops its pair ([FD] §3.2.2).

**DEC2** 2026-10-07: 750-455 word layout confirmed by WAGO (value in bits 14..4, 0..2047).

**DEC3** 2026-10-15 (PLC, in review): no valid pair ⇒ alarm LevelMeasurementLost and LIC01 in manual at SafeOutput (60 rpm), bumpless return.

**DEC4** 2026-10-13: LIC01 start value $T_i$ = 60 s instead of 600 s, from the design basis and the simulator test; to be confirmed at commissioning.

**DEC5** 2026-10-22: LIC01's output is the total speed of the pumps in level control; the pumps in REMOTE share it after the actual speed of the others in level control. $K_p$ 450 per pump → 1350 total; per-pump limits, `SafeOutput` and manual output scaled by $N$ (IN-05, IN-07).

**DEC6** 2026-10-22: InletRIO $n$ link lost ⇒ both transmitters of inlet $n$ bad, its pair drops out of the PV (IN-19).

**DEC7** 2026-10-22: automatic pump count and rotation (Section 3.4): candidates = Enable, auto, not UNAVAILABLE / LOCKDOWN; start all at Pen Start, then reduce; settings retained in `PenConfig` (IN-15).

**DEC8** 2026-10-22: level deviation alarms high / low with LIC01 in auto, limit and delay retained in `PenConfig` (IN-09).